Conclusions for assignment_1

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('M3', 'Simulator', 'A') median Tm = 0.0148 s
```

	PWM	omega_ss_rad_s	Tm_s	omega_se	Tm_se	angle_rmse_rad
0	-100	-9.6004	0.0159	0.0063	0.0008	0.0496
1	-90	-8.6409	0.0160	0.0058	0.0008	0.0451
2	-80	-7.6801	0.0158	0.0051	0.0008	0.0399
3	-70	-6.7205	0.0163	0.0045	0.0008	0.0352
4	-60	-5.7604	0.0165	0.0039	0.0008	0.0303
5	-50	-4.8000	0.0167	0.0032	0.0008	0.0252

The main goal from these columns was:

1)How fast does the motor rotates at particular PWM values
2)How quickly does the motor reach that speed (Tm)

Omega_ss_rad_s is steady state angular speed of the motor.

w= change in theta over time

Tm = Time taken to reach the 63.2 percent of the final speed

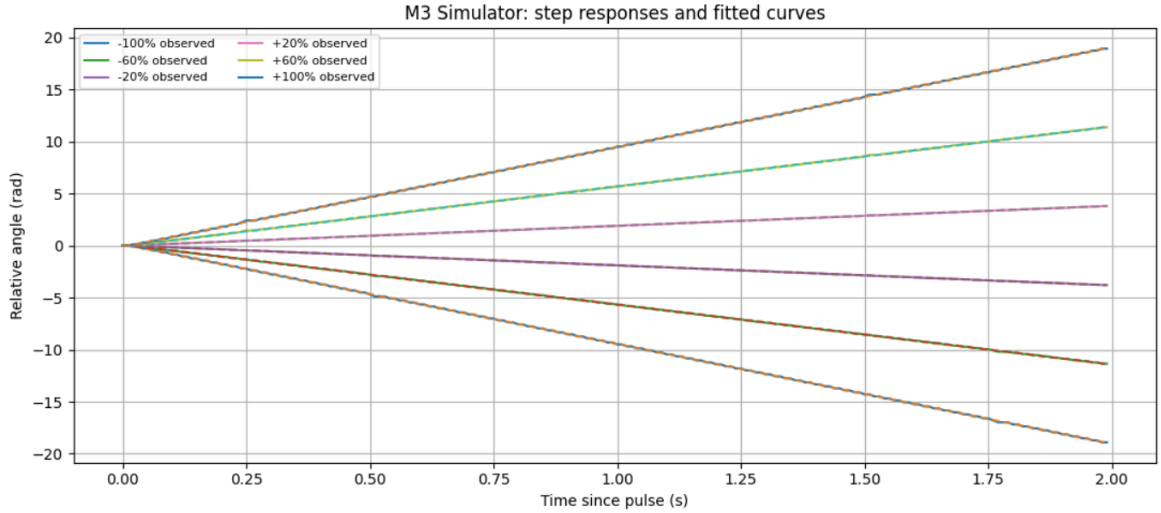
T□(dω/dt) + ω = K × PWM

Omega_se = Standard error of angular velocity.Estimated uncertainty in that speed.

Tm_se = standard error of the motor time constant.Estimated uncertainty in that motor time constant.

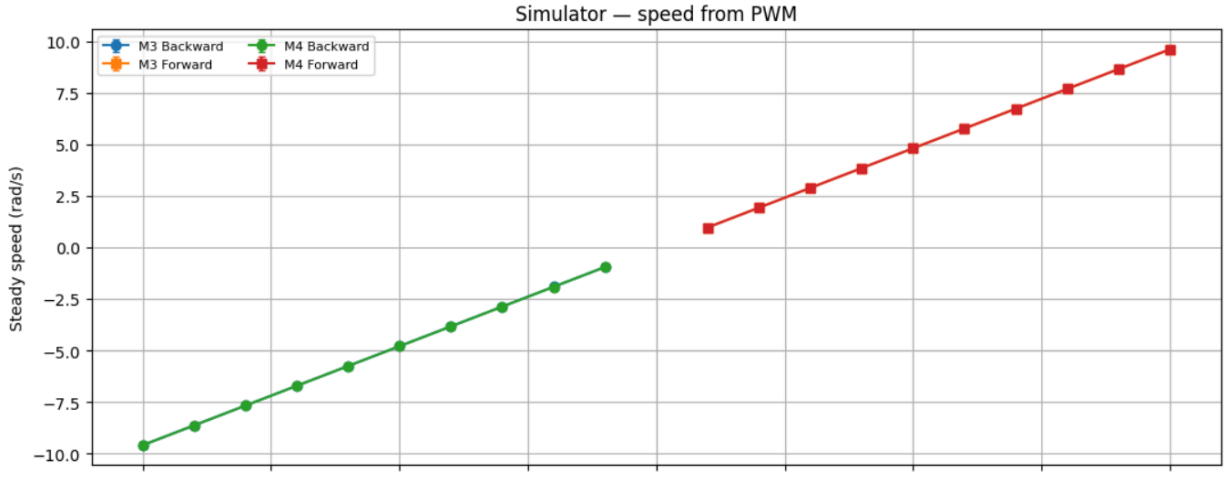
angle_RMSE measures the difference between the **encoder angles predicted by our mathematical model** and the angles actually recorded during the experiment.

θ(t) = θ₀ + ω□□[t − T□(1 − e^(−t/T□))] - predicted angle by mathematical model



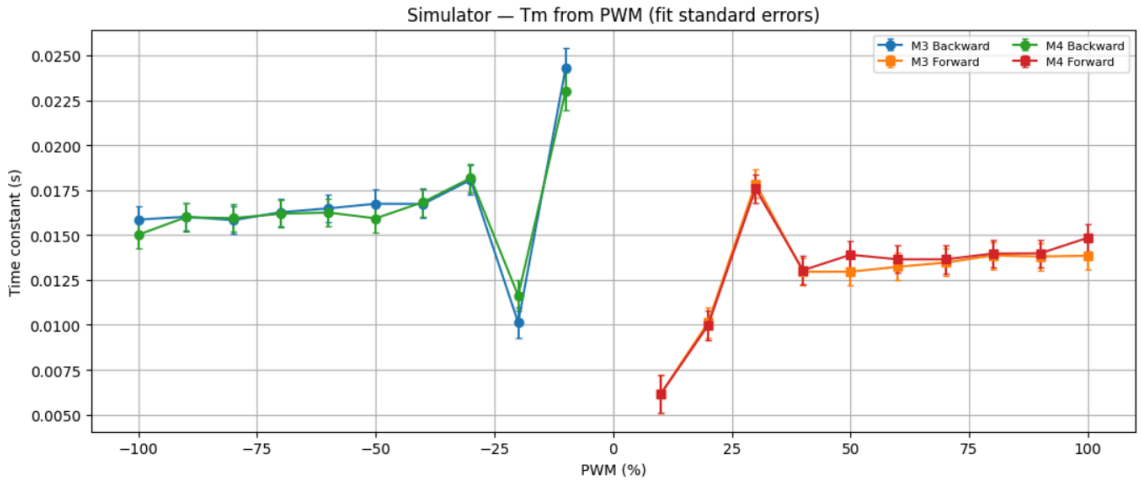
This graph checks how accurately our mathematical model predicts the motor's movement at different PWM values.

Our graph shows how encoder angle changes when we change the PWM values. So it shows motor speed is directly proportional to PWM values.

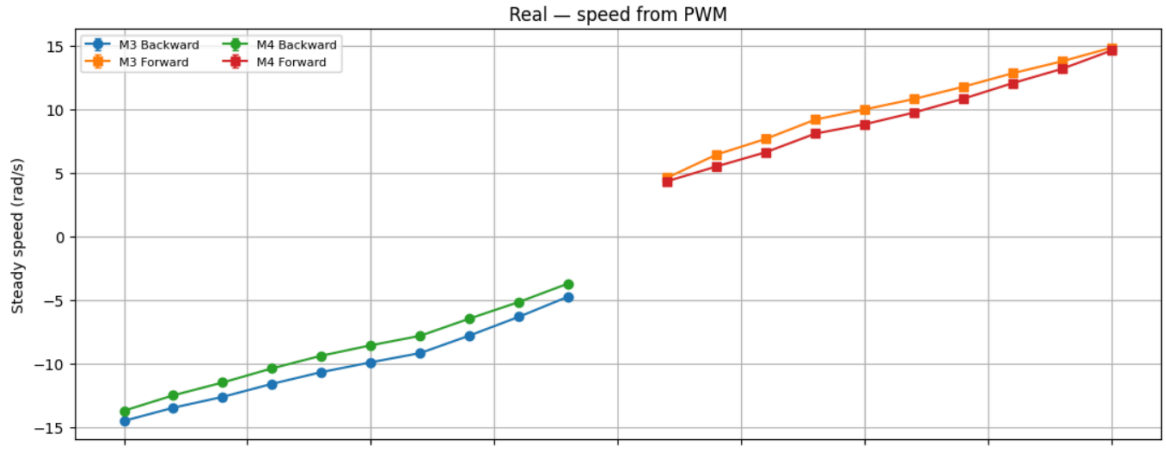


The graph shows a linear relationship between PWM and motor speed. As PWM increases, motor speed increases proportionally. Positive PWM produces forward rotation, while negative PWM

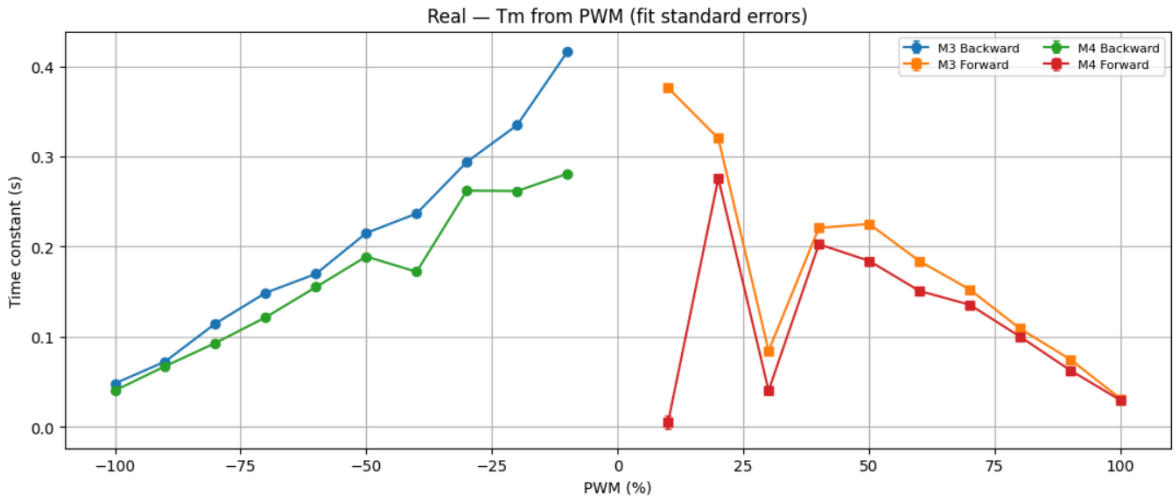
produces backward rotation. Both motors show similar behavior in the simulator.



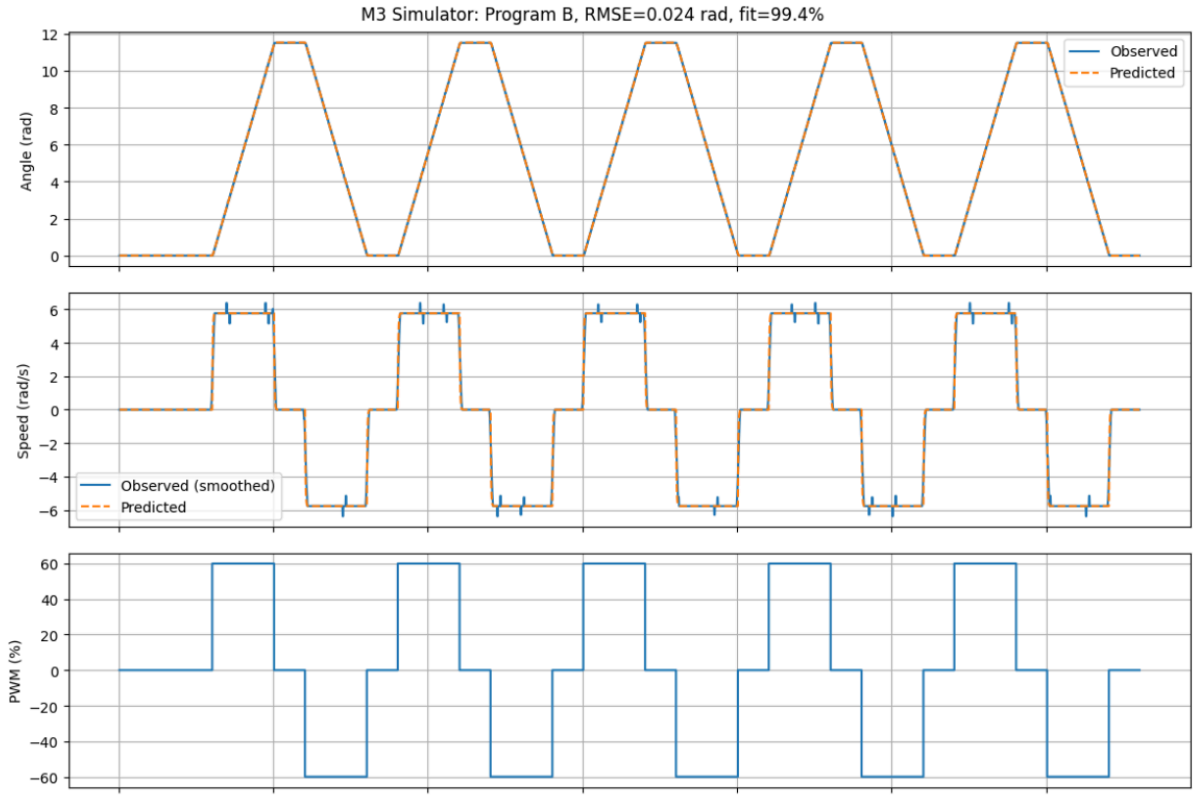
The motor time constant remains mostly stable across PWM values, with some fluctuations at low PWM . Both motors show similar behavior. At low PWM values Tm increases it means at low motor power the robot takes a bit more time to respond to the power and reach its steady-state speed.



Motor speed increases almost linearly with PWM. M3 and M4 show slight speed differences, indicating real-world motor variations.

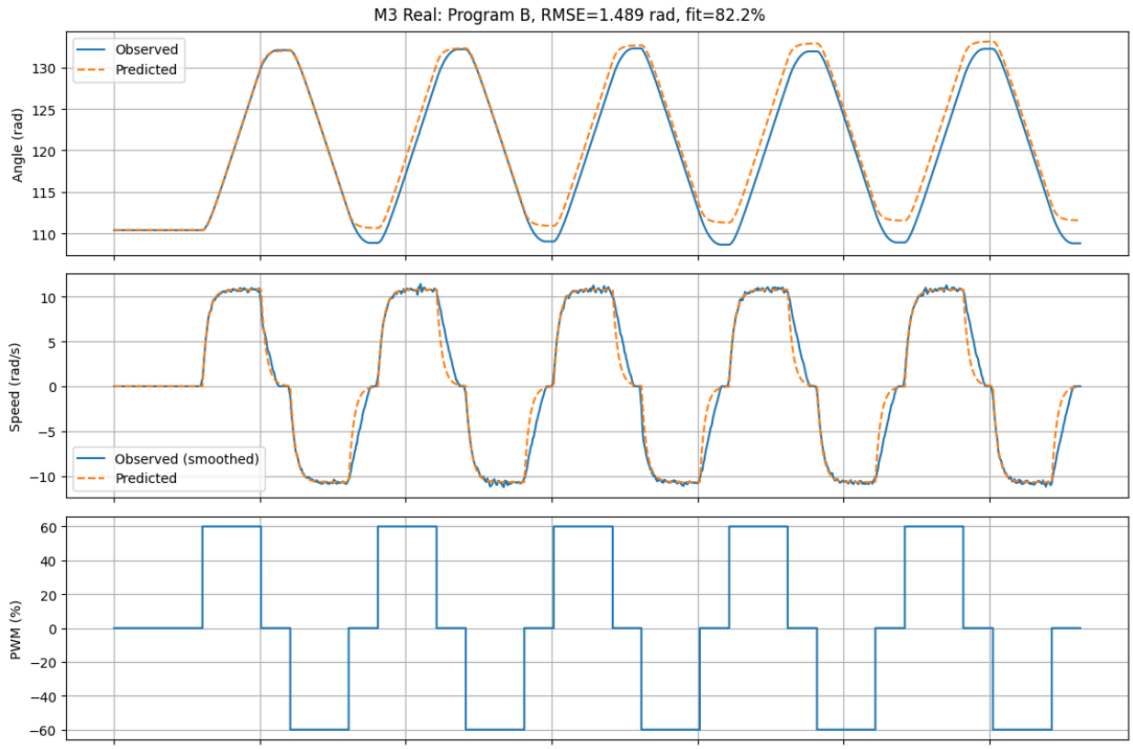


The motor time constant generally decreases at higher PWM magnitudes(making the robot response faster), with noticeable differences between M3 and M4.

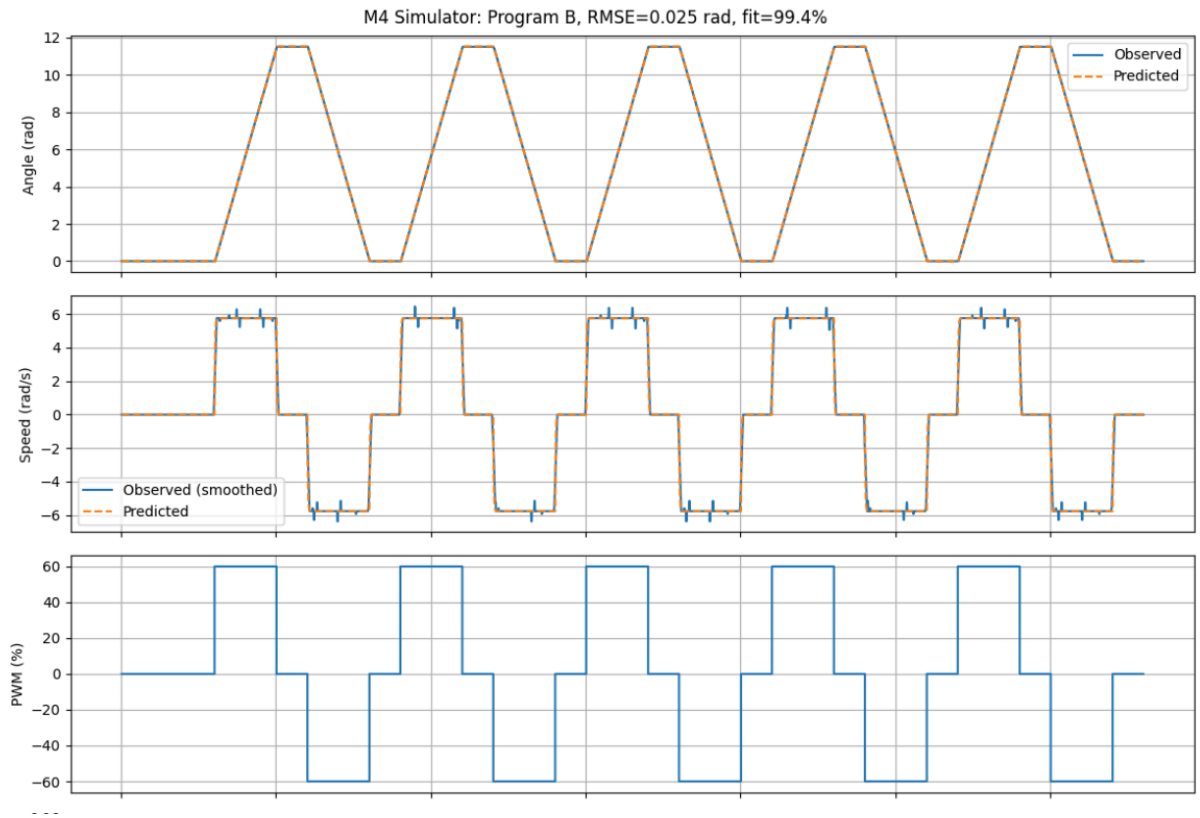


The predicted and observed motor responses match closely, with,

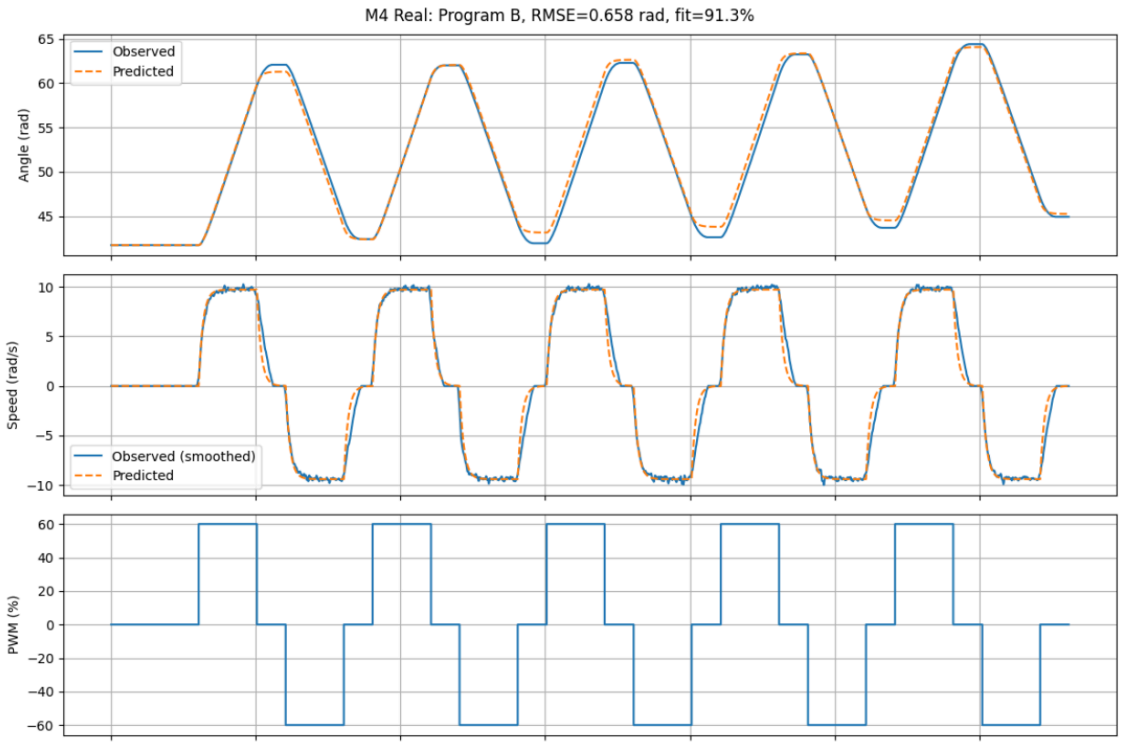
a 99.4% fit and low RMSE (0.024 rad), indicating high model accuracy in the simulator.



The model shows a good fit of 82.2% with RMSE = 1.489 rad, although real motor behavior differs slightly from predictions.



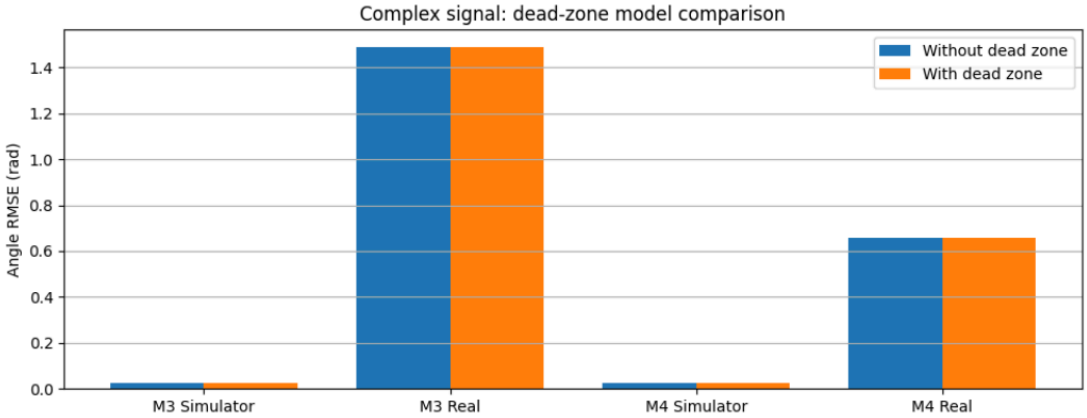
The predicted and observed M4 responses match almost perfectly, with a 99.4% fit and a low RMSE of 0.025 rad.



The predicted and observed M4 responses match closely, with a 91.3% fit and RMSE of 0.658 rad, indicating good model accuracy on the real robot.

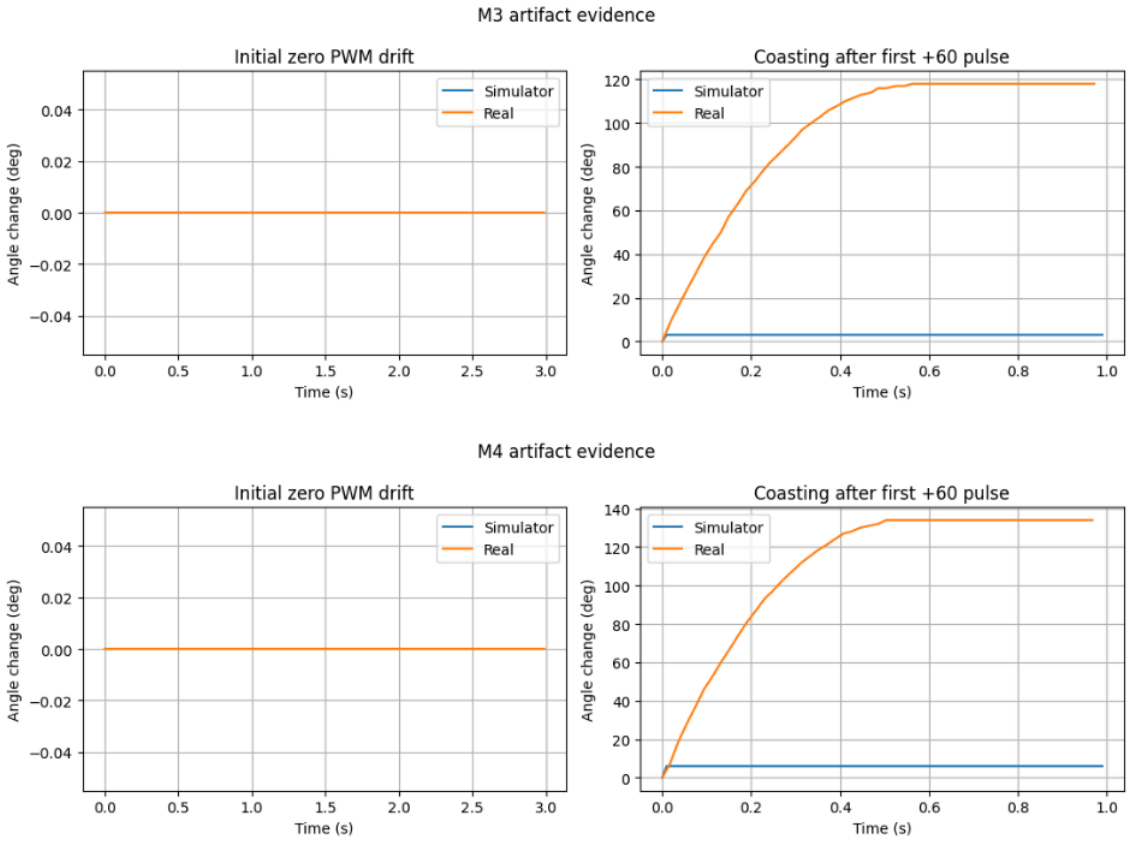
	Motor	Environment	Tm_median_s	angle_RMSE_rad	angle_fit_percent	speed_smoothing_samples
0	M3	Simulator	0.0148	0.0244	99.4471	10
1	M3	Real	0.1769	1.4889	82.1692	5
2	M4	Simulator	0.0149	0.0247	99.4403	10
3	M4	Real	0.1432	0.6578	91.3092	5

	Motor	Environment	indicative_deadzone_PWM	RMSE_without_rad	RMSE_with_rad
0	M3	Simulator	0.0013	0.0244	0.0244
1	M3	Real	20.0000	1.4889	1.4889
2	M4	Simulator	0.0034	0.0247	0.0247
3	M4	Real	20.0000	0.6578	0.6578

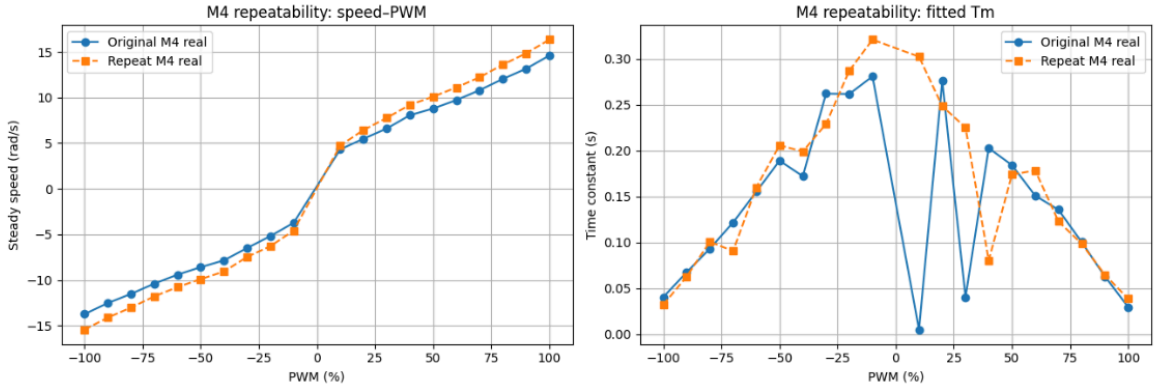


The dead-zone model does not reduce RMSE, and simulation errors remain lower in simulation than real-world errors.

	Motor	Environment	initial_zero_drift_deg	first_stop_coast_deg	median_dt_ms	duplicate_timestamps
0	M3	Simulator	0.0	3.0	10.0	10
1	M3	Real	0.0	118.0	20.0	0
2	M4	Simulator	0.0	6.0	10.0	10
3	M4	Real	0.0	134.0	20.0	0

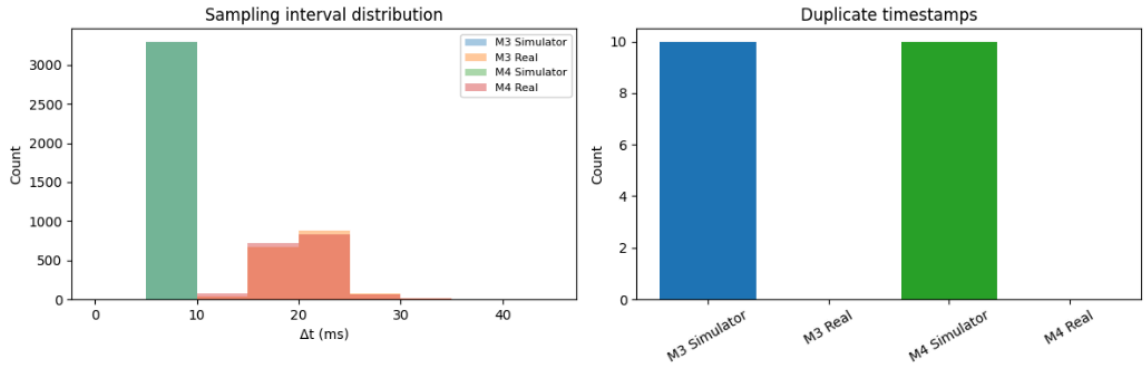


Both motors show no initial drift, but the real motors exhibit significantly more drifting than in the simulator.



M4 speed is consistent, but its time constant varies between

repeated same tests. The motor takes different amounts of time to reach 63.2% of its final speed in the two experiments.



The simulator collects data faster but has some duplicate timestamps, while the real robot collects data more slowly without duplicates.

In this experiment, we studied how motors M3 and M4 respond to different PWM values. We found that motor speed increases almost linearly with PWM and that the motors respond faster in the simulator than on the real robot. Our mathematical model worked very well in the simulator, but the real motors showed some differences due to physical factors such as friction. Overall, this experiment helped us better understand motor behavior and why testing on a real robot is important.

Additional conclusions

Motor model assumptions

The model assumes constant battery voltage and load, simple friction, and ignores motor inductance. These assumptions work well in the simulator, but real motors have more variation.

Forward and backward rotation

Positive and negative PWM make the motor turn in opposite directions. The real motors do not respond exactly the same way in both directions, so separate settings could improve control.

What matters for odometry

Coasting and differences between M3 and M4 can make the robot travel farther or turn differently than expected. Noisy encoder readings can also affect distance estimates, so these errors matter for odometry.

Where the model makes mistakes

The biggest differences appear around changes in direction, when the motor slows down and starts moving the other way. The model still follows the overall movement well, especially in the simulator.